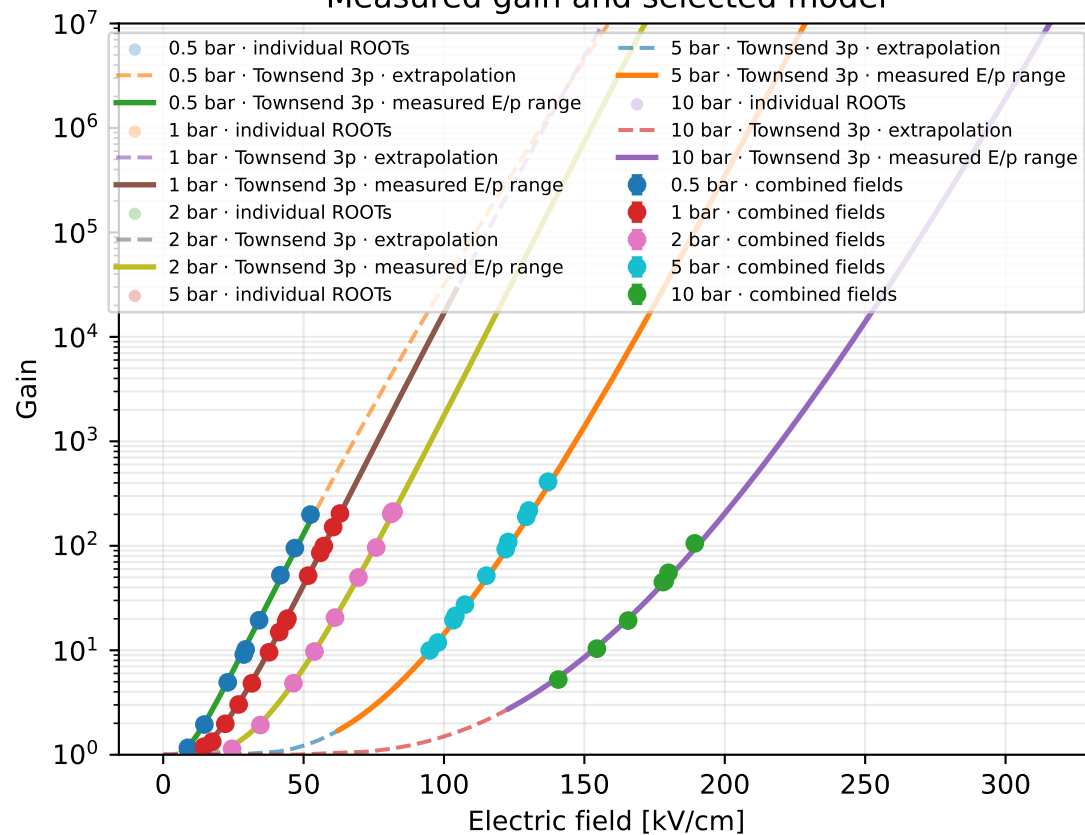
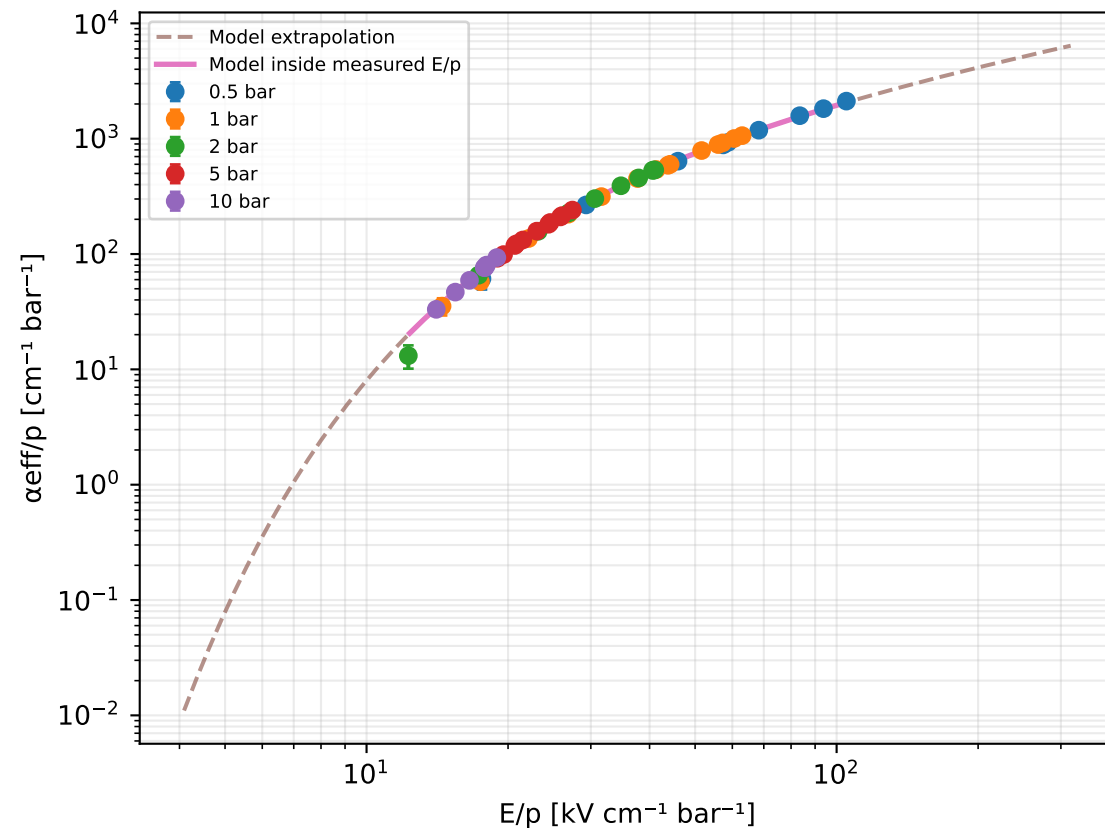


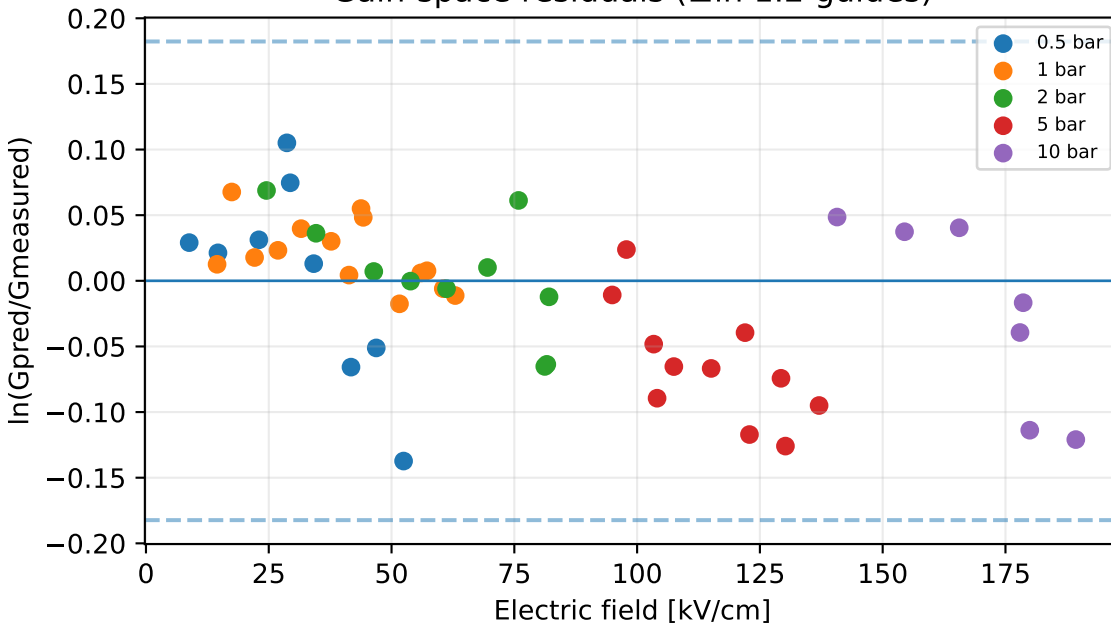
Measured gain and selected model



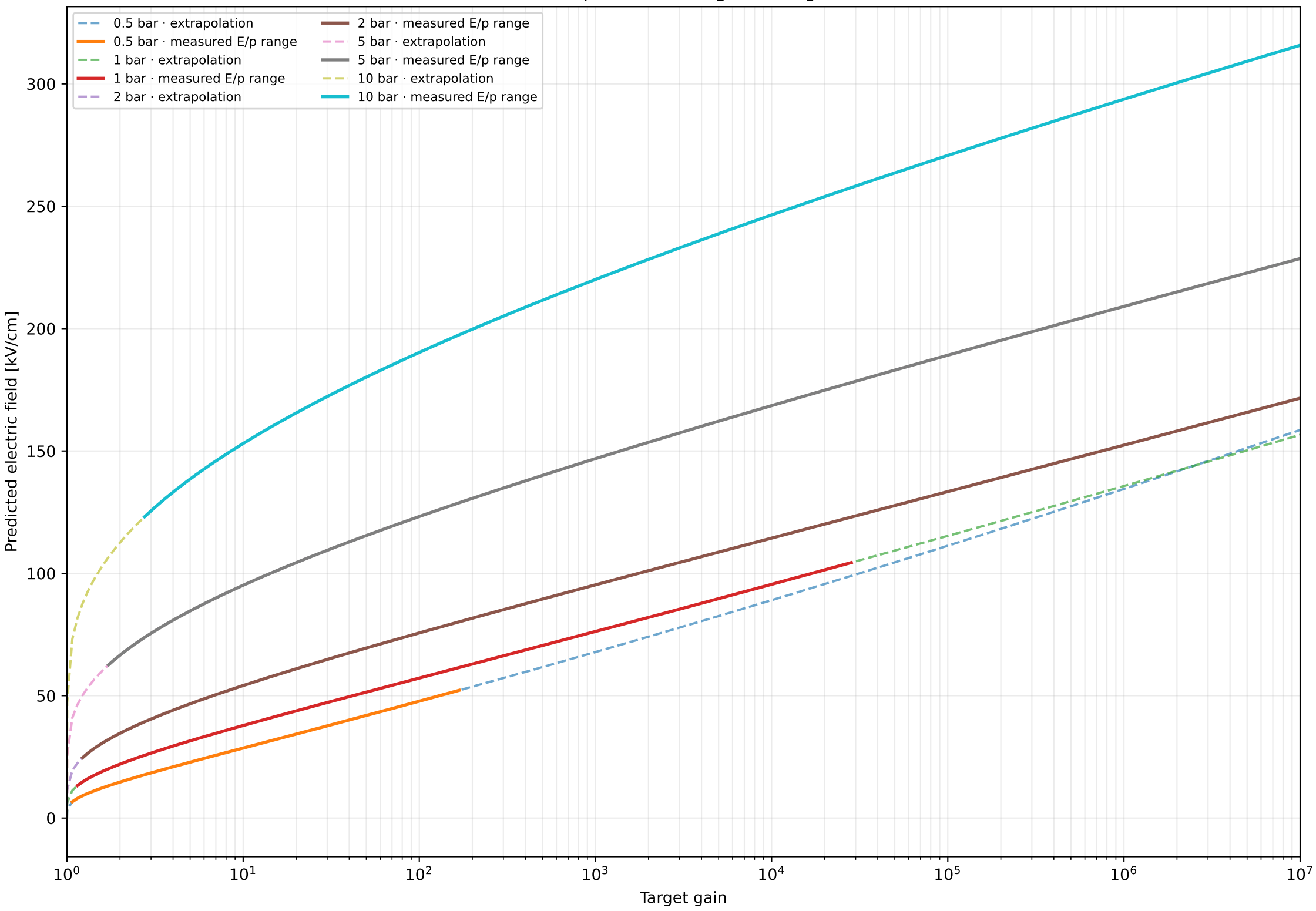
Reduced Townsend fit



Gain-space residuals ($\pm \ln 1.2$ guides)



ROOT runs: 73
 Unique physical points: 51
 Pressures: 0.5, 1, 2, 5, 10 bar
 Selected model: Townsend 3p
 Status: accepted
 Training RMSE: $\times 1.076$
 Cross-validation median: $\times 1.043$
 Cross-validation worst: $\times 1.218$
 Reduced-field range: 12.27–104.9 $\text{kV cm}^{-1} \text{bar}^{-1}$
 $A=74.994$, $B=40.685$, $m=0.79561$, $n=1$
 Covariance condition: $2.53\text{e}+04$

Field extrapolation from gain 1 to gain 10^7 

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm ⁻¹]	npe	runs
0.5	8.80059	17.6012	1.165217	0.0348	30.5815	115	1
0.5	14.6573	29.3146	1.947368	0.0583	133.296	361	1
0.5	22.9924	45.9848	4.934641	0.148	319.256	459	1
0.5	28.6899	57.3799	9.103723	0.273	441.737	376	1
0.5	29.369	58.738	10.19134	0.306	464.308	439	1
0.5	34.152	68.3041	19.37532	0.581	592.8	389	1
0.5	41.7428	83.4856	52.34294	1.57	791.563	347	1
0.5	46.8862	93.7723	95.0669	2.84	910.916	284	1
0.5	52.453	104.906	198.9933	6.08	1058.65	300	1
1	14.4769	14.4769	1.193182	0.0358	35.3247	176	1
1	17.4658	17.4658	1.333333	0.04	57.5364	285	1
1	22.1139	22.1139	1.977427	0.0593	136.359	443	1
1	26.8704	26.8704	3.028517	0.0908	221.615	526	1
1	31.5919	31.5919	4.825046	0.145	314.764	543	1
1	37.7055	37.7055	9.570076	0.287	451.728	528	1
1	41.3391	41.3391	14.92692	0.447	540.633	520	1
1	43.7888	43.7888	18.91225	0.567	587.962	547	1
1	44.2397	44.2397	20.08052	0.601	599.95	534	1
1	51.6157	51.6157	51.80328	1.55	789.491	488	1
1	55.9603	55.9603	85.57333	3.38	889.875	300	1
1	57.2166	57.2166	99.4952	2.98	920.022	521	1
1	60.5352	60.5352	150.9133	5.6	1003.34	300	1
1	62.9584	62.9584	203.64	7.8	1063.27	300	1
2	24.5426	12.2713	1.140426	0.0342	26.2803	235	1
2	34.6263	17.3132	1.925512	0.0577	131.038	537	1
2	46.3843	23.1922	4.834586	0.145	315.159	665	1
2	53.8679	26.9339	9.729899	0.292	455.041	796	1
2	61.1917	30.5958	20.54762	0.615	604.549	630	1
2	69.5334	34.7667	49.75195	1.49	781.41	641	1
2	75.8577	37.9288	96.26404	2.88	913.419	534	1

p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
2	81.2007	40.6004	202.03	7.72	1061.68	300	1
2	81.6124	40.8062	211.59	7.97	1070.93	300	1
2	82.0652	41.0326	211.8467	9.02	1071.17	300	1
5	94.9317	18.9863	9.9637	0.299	459.79	854	1
5	97.8221	19.5644	11.8554	0.355	494.557	982	1
5	103.378	20.6755	19.40382	0.582	593.094	889	1
5	104.066	20.8132	21.33893	0.64	612.107	894	1
5	107.478	21.4956	27.35301	0.82	661.765	830	1
5	115.078	23.0157	51.80321	1.55	789.49	747	1
5	121.976	24.3953	92.89667	3.32	906.298	600	1
5	122.877	24.5755	108.97	3.87	938.215	600	1
5	129.295	25.859	189.6567	6.62	1049.04	600	1
5	130.199	26.0398	217.62	10.4	1076.55	300	1
5	137.053	27.4105	409.85	6.1	1203.16	3000	10
10	140.714	14.0714	5.245192	0.157	331.462	1248	1
10	154.448	15.4448	10.36355	0.311	467.659	1092	1
10	165.585	16.5585	19.24643	0.577	591.465	1120	1
10	177.961	17.7961	44.95214	0.463	761.12	8400	14
10	178.604	17.8604	45.8296	1.37	764.986	1027	1
10	179.948	17.9948	55.18067	1.65	802.123	869	1
10	189.32	18.932	105.5158	3.17	931.772	952	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-223	×1.19	×6.03	no	large cross validation outlier
Townsend 3p	3	-260	×1.04	×1.22	yes	—
Townsend 4p	4	-273	×1.03	×1.1	yes	—